

Employment

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| <b>Graduate Research Assistant</b>                                                                                                                                                                                                                                                                                                                                                               | <b>NC State University</b>  | <b>Apr 2025 – Present</b>  |
| <ul style="list-style-type: none"><li>Collaborated with plant biologists and geneticists to deploy cloud-native genomic data platform on GCP (Cloud Run) with JBrowse 2 genome browser, enabling 100+ researchers to visualize 500GB+ genome datasets.</li><li>Improved ML model accuracy by 12% via hyperparameter tuning and time-series feature analysis on biological sensor data.</li></ul> |                             |                            |
| <b>Product Engineer Intern</b>                                                                                                                                                                                                                                                                                                                                                                   | <b>Isomer AI</b>            | <b>May 2025 – May 2026</b> |
| <ul style="list-style-type: none"><li>Shipped per-tenant AI workflow platform for insurance carriers: Temporal.io durable orchestration, LLM risk signal detection across Claude/GPT-4o/Gemini, Auth0 RBAC, MCP servers; 30+ deployments.</li><li>Built MCP servers (Python + TypeScript); Playwright + Tailscale VPN for secure customer-internal browser automation.</li></ul>                 |                             |                            |
| <b>Software Developer Intern</b>                                                                                                                                                                                                                                                                                                                                                                 | <b>Mphasis (HP Company)</b> | <b>Jun 2024 – Aug 2024</b> |
| <ul style="list-style-type: none"><li>Partnered with healthcare data teams to build 5+ AI workflows including health claim automation using OCR, NLP, and ML pipelines; improved document processing accuracy by 25% across 10K+ daily medical claims.</li><li>Automated AWS and GCP CI/CD pipelines; integrated Grafana monitoring and led model validation and red-teaming sessions.</li></ul> |                             |                            |
| <b>AI/ML Intern</b>                                                                                                                                                                                                                                                                                                                                                                              | <b>Capgemini</b>            | <b>Jun 2023 – Aug 2023</b> |
| <ul style="list-style-type: none"><li>Led cross-functional SDLC for data-driven enterprise ticket analysis using NLP classification models; worked with support teams to reduce resolution time by 15% and eliminate 2,000+ monthly manual triages.</li><li>Built Kafka + Spark + Hadoop pipelines with data engineers; cut ticket-processing latency by 20% at peak daily load.</li></ul>       |                             |                            |

Education

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| <b>Raleigh, NC</b>                                                                                                                                                                                                                                                                                    | <b>North Carolina State University</b> | <b>Aug 2024 – May 2026</b> |
| <ul style="list-style-type: none"><li>Master of Science in Computer Science; GPA: 3.5/4.0. Courses: Artificial Intelligence, Operating Systems, Algorithms, Data Science, Object Oriented Development, Software Engineering, Programming Languages, Computer Networks, Distributed Systems.</li></ul> |                                        |                            |
| <b>Mumbai, India</b>                                                                                                                                                                                                                                                                                  | <b>Mumbai University</b>               | <b>Aug 2020 – May 2024</b> |
| <ul style="list-style-type: none"><li>Bachelor of Technology in Computer Science; CGPA: 9.32/10. Courses: Data Warehousing, Big Data Infrastructure, DS &amp; Algorithms, Machine Learning, Blockchain, Computer Networks, Database Systems.</li></ul>                                                |                                        |                            |

Projects

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| <b>LLM Eval Harness</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>Vendor-Neutral LLM Benchmark</b>      | <b>2025</b> |
| <ul style="list-style-type: none"><li>Built vendor-neutral LLM evaluation harness in Python comparing Claude, GPT-4o, and Gemini across 2,000 frozen labeled examples spanning factual recall, multi-step reasoning, and code generation tasks.</li><li>Tracked hallucination rate, p50/p95 latency, and cost-per-call; applied McNemar significance tests across 6 metrics.</li></ul>                                                                                                                        |                                          |             |
| <b>KubeServe</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Kubernetes LLM Inference Platform</b> | <b>2025</b> |
| <ul style="list-style-type: none"><li>Built Kubernetes-native LLM inference platform: FastAPI router, KEDA autoscaler (scale-to-zero to N replicas), Kopf CRD operator for multi-provider LLM serving; circuit breaker + EWMA load balancing.</li><li>Deployed via Helm + ArgoCD GitOps on Kubernetes; OTel + Prometheus + Grafana observability stack; 88 pytest tests.</li></ul>                                                                                                                            |                                          |             |
| <b>AlgoPulse</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Full-Stack Agentic AI Platform</b>    | <b>2025</b> |
| <ul style="list-style-type: none"><li>Built agentic AI orchestration platform on Temporal.io and Kafka: agent loop with multi-step reasoning, tool invocation, and retry logic; sustained 2,000 async inference requests per second.</li><li>Deployed on AWS with auto-scaling groups; k6 load tested to 10,000 concurrent users; sub-100ms leaderboard latency.</li><li>Designed event-driven SSE delivery via Supabase Edge Functions; replaced polling and cut active server connections by 60%.</li></ul> |                                          |             |

Languages and Technologies

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| <ul style="list-style-type: none"><li><b>Languages:</b> Python, TypeScript, JavaScript, Java, SQL, Bash</li><li><b>AI &amp; Agentic:</b> LLM Evals, Claude API, GPT-4o, Gemini, MCP, Temporal.io, Kafka, RAG, LangChain, Prompt Engineering</li><li><b>Infra &amp; Observability:</b> Kubernetes, Docker, FastAPI, GCP, AWS, OpenTelemetry, Prometheus, Grafana, GitHub Actions</li><li><b>Data &amp; Backend:</b> PostgreSQL, Redis, MongoDB, DuckDB, Parquet, Terraform, Helm, ArgoCD, Testcontainers, k6</li></ul> |  |  |
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Additional Experience and Awards

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| <ul style="list-style-type: none"><li><b>IEEE Published:</b> Co-authored and presented research on Computer Vision-Based Cybersecurity Threat Detection at the Springer icSoftComp 2023 international conference, reaching an audience of 500+ global attendees.</li><li><b>Hackathon Winner:</b> Won or ranked Top 10 in 6+ national hackathons; built 3 full-stack MVPs end-to-end in 48-hour competitive sprints.</li><li><b>GDSC Lead:</b> Led 4+ tech events as Google Developer Student Club Lead, reaching 500+ students across hackathons and conferences.</li><li><b>Technical Writing:</b> Published articles on LLM model collapse, AI agent security, and audio AI at kulkarniaditya.com/blog.</li></ul> |  |  |
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